SymPy Bug Card

Bug 47: linsolve misses the a = 0 branch

Task. Solve the symbolic-parameter linear equation for x:

$$ax - a = 0.$$

SymPy's answer.

$$\operatorname{linsolve}([ax - a], [x]) = \{(1,)\}.$$

Correct answer.

$$a \neq 0: \; x = 1, \qquad a = 0: \; x \text{ arbitrary}.$$

Explanation. The equation factors as $a(x-1) = 0$. When $a = 0$, it becomes $0 = 0$, so $x = 2$ is a valid solution; after substituting $a = 0$, SymPy's reported set still contains only $(1,)$.

Likely root cause. The diagnosis identifies `sympy/polys/matrices/linsolve.py`: the sparse linear solver builds the coefficient domain as $\mathbb{Z}(a)$, where a is treated as invertible, so RREF divides by a and returns only the generic branch. This is an instance of the known symbolic-coefficient missing-$a = 0$-branch problem tracked in open issue sympy/sympy#16861.

Artifact (GitHub). `bug-report/bug-047-linsolve-misses-the-a-0-branch/`